Introduction to Gravitational Waves and Gravitational Waves Observation

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Contents

1 Introduction

Gravitational waves, distortions in spacetime caused by massive accelerating bodies, stand among the most profound discoveries in contemporary physics. First predicted by Albert Einstein in 1916 as a consequence of his General Theory of Relativity, these waves remained undetected for nearly a century. That changed in 2015, when the Laser Interferometer Gravitational-Wave Observatory (LIGO) made the first direct observation—confirming Einstein's prediction and inaugurating a new era in observational astronomy.

This project investigates the essential nature of gravitational waves, the astrophysical processes that generate them, and the advanced technologies designed to detect them.

2 General Relativity and the Origin of Gravitational Waves

The aim of this section is to outline the fundamental principles of General Relativity, the theoretical framework from which gravitational waves were first predicted.

2.1 What is a Gravitational Wave?

A gravitational wave (GW) is any tensorial perturbation in the curvature of spacetime that produces oscillatory effects. These fluctuations propagate through empty space at the speed of light.

In general, there are two main challenges in the theoretical treatment of this phenomenon.

- **Nonlinearity of Einstein's Equations:** The Einstein field equations are nonlinear, which means we can only study approximate solutions [1]. As a result, gravitational waves cannot, in general, be superimposed like linear waves. In this project, we focus on a theory of weak perturbations in flat spacetime, which is typically a good approximation for the conditions under which gravitational waves propagate in astrophysical settings.

- **Gauge Freedom in General Relativity:** General Relativity allows for a large degree of gauge freedom. This means we must carefully distinguish whether the oscillatory effects observed in a given metric are due to actual physical perturbations in the geometry of spacetime, or if they are simply artifacts of the coordinate system chosen.[2]

[1] This holds except in cases with specific symmetries, such as the Schwarzschild solution.

[2] An example on how the Gauge Freedom can be a problem is given in the section 2.3

2.2 Linearised Gravity

As discussed earlier, we consider a flat spacetime subject to small perturbations. Under these conditions, we aim to find a solution to Einstein's equations in vacuum. Such a solution will describe how the perturbation propagates through empty space. Since we have already made certain assumptions, we can adopt the following metric as an ansatz:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}(x) \tag{1}$$

Here, $\eta_{\mu\nu}$ is the Minkowski metric, chosen to reflect the assumption of flat spacetime. Adopting the East Coast convention commonly used in General Relativity, it is given by:

$$\begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The quantity $h_{\mu\nu}$ represents a small tensor perturbation. Since both $g_{\mu\nu}$ and $\eta_{\mu\nu}$ are symmetric metrics, $h_{\mu\nu}$ must also be symmetric.

Under these assumptions, the Christoffel symbols take the form:

$$\Gamma^{\mu}_{\nu\rho} = \frac{1}{2} g^{\mu\sigma} \left(\partial_\nu g_{\sigma\rho} + \partial_\rho g_{\sigma\nu} - \partial_\sigma g_{\nu\rho} \right) \tag{2}$$

To evaluate this, we first need the inverse metric $g^{\mu\nu}$, which satisfies:

$$\delta_{\mu}{}^{\sigma} = g_{\mu\nu} g^{\nu\sigma} = (\eta_{\mu\nu} + h_{\mu\nu})(\eta^{\nu\sigma} + k^{\nu\sigma}) = \eta_{\mu\nu}\eta^{\nu\sigma} + \eta_{\mu\nu}k^{\nu\sigma} + h_{\mu\nu}\eta^{\nu\sigma} + h_{\mu\nu}k^{\nu\sigma} \tag{3}$$

The last term is second order in the perturbation and can be neglected, while the first contraction obviously yields $\delta_{\mu}{}^{\sigma}$. Thus, to first order, the inverse metric becomes:

$$g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu} + \mathcal{O}(h^2) \tag{4}$$

Substituting this back into the Christoffel symbols gives:

$$\Gamma^{\mu}_{\nu\rho} = \frac{1}{2} \eta^{\mu\sigma} \left(\partial_\nu h_{\sigma\rho} + \partial_\rho h_{\sigma\nu} - \partial_\sigma h_{\nu\rho} \right) + \mathcal{O}(h^2) \tag{5}$$

The Riemann tensor is then:

$$R^{\mu}{}_{\nu\sigma\rho} = \partial_\sigma \Gamma^{\mu}_{\nu\rho} - \partial_\rho \Gamma^{\mu}_{\nu\sigma} + \Gamma^{\mu}_{\sigma\lambda}\Gamma^{\lambda}_{\rho\nu} - \Gamma^{\mu}_{\rho\lambda}\Gamma^{\lambda}_{\sigma\nu} = \partial_\sigma \Gamma^{\mu}_{\nu\rho} - \partial_\rho \Gamma^{\mu}_{\nu\sigma} + \mathcal{O}(h^2) \tag{6}$$

From this, the Ricci tensor follows:

$$R_{\nu\rho} = R^{\mu}{}_{\nu\mu\rho} = \partial_\mu \Gamma^{\mu}_{\nu\rho} - \partial_\rho \Gamma^{\mu}_{\nu\mu} + \mathcal{O}(h^2) \tag{7}$$

$$= \tfrac{1}{2}\eta^{\mu\sigma}\partial_\mu\big(\partial_\rho h_{\sigma\nu} + \partial_\nu h_{\sigma\rho} - \partial_\sigma h_{\nu\rho}\big) - \tfrac{1}{2}\eta^{\mu\sigma}\partial_\rho\big(\partial_\nu h_{\sigma\mu} + \partial_\mu h_{\sigma\nu} - \partial_\sigma h_{\nu\mu}\big) \tag{8}$$

$$= \frac{1}{2}\Big(\partial^\sigma\partial_\nu h_{\sigma\rho} + \partial^\sigma\partial_\rho h_{\sigma\nu} - \partial^\sigma\partial_\sigma h_{\nu\rho} - \partial_\nu\partial_\rho h + \partial_\rho\partial^\sigma h_{\sigma\rho} - \partial_\rho\partial^\sigma h_{\sigma\rho}\Big) \tag{9}$$

$$= \tfrac{1}{2}\Big(\partial^\sigma\partial_\nu h_{\sigma\rho} + \partial^\sigma\partial_\rho h_{\sigma\nu} - \Box h_{\nu\rho} - \partial_\nu\partial_\rho h\Big), \tag{10}$$

where $\Box = g^{\mu\nu}\partial_\mu\partial_\nu$ is the d'Alembertian, and $h = h^{\mu}{}_{\mu}$ is the trace.
In vacuum, the Einstein equations reduce to:

$$R_{\mu\nu} = 0 \tag{11}$$

Using the linearised form, we obtain:

$$\Box h_{\mu\nu} + \partial_\mu\partial_\nu h - \partial_\mu\partial^\sigma h_{\sigma\nu} - \partial_\nu\partial^\sigma h_{\sigma\mu} = 0 \tag{12}$$

We now define the vector:

$$V_\mu := \partial_\nu h^{\nu}{}_{\mu} - \tfrac{1}{2}\partial_\mu h \tag{13}$$

Since spacetime is nearly flat, we can use the Minkowski metric to raise and lower indices as shown:

$$\begin{aligned}
\partial_\sigma h^{\sigma}{}_{\mu} &= \partial_\sigma\big(g^{\sigma\rho}h_{\rho\mu}\big) &(14)\\
&= \partial_\sigma\big(\eta^{\sigma\rho}h_{\rho\mu} + \mathcal{O}(h^2)\big) &(15)\\
&= \eta^{\rho\sigma}\partial_\sigma h_{\rho\mu} + \mathcal{O}(h^2) &(16)\\
&= \partial^\rho h_{\rho\mu} + \mathcal{O}(h^2) &(17)
\end{aligned}$$

The linearised equation can then be written as:

$$\Box h_{\mu\nu} = \partial_\mu V_\nu + \partial_\nu V_\mu \tag{18}$$

At this stage, the wave-like nature of the perturbation tensor becomes apparent. For a free wave, we set the d'Alembertian to zero. It can be shown that a suitable gauge transformation exists such that $V_\mu = 0$ (the proof is omitted here, as it is lengthy and not essential for our purposes).

In this gauge, the equation reduces to:

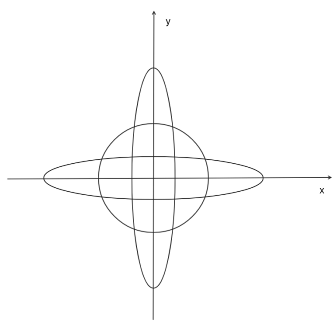
$$\Box h_{\mu\nu} = 0 \quad \forall\, \mu,\nu \tag{19}$$

2.3 Polarisation

The perturbation tensor $h_{\mu\nu}$ is a symmetric tensor (as previously shown) that also has other particular features. Without going into the details of the demonstration, it can be shown that, in general, the matrix for a wave propagating along the z-axis is built as follows:

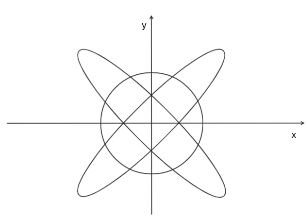
$$h = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & a_+ & a_\times & 0 \\ 0 & a_\times & -a_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

This means that the perturbation tensor can be expressed as the linear combination of two other tensors:



$$h_+ = a_+ \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Plus polarisation (+)



$$h_\times = a_\times \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Cross polarisation (×)

The two independent patterns above correspond to the deformation that a circle of free-falling masses, perpendicular to the direction of propagation recives as it is passed by pure *plus* (+) or *cross* (×) polarisations of gravitational waves. These represent the two fundamental modes of distortion that a gravitational wave induces in spacetime as it passes through.

2.4 Example 1: How Can Gauge Freedom Be Misleading?

In principle, one could develop a theory of gravitational waves with some sort of gauge restriction that restricts problematic coordinate transformations (In particular in linearized theory it's common to choose the Lorenz gauge). However, since we are working within the framework of General Relativity—which allows arbitrary coordinate choices—we can construct examples where a coordinate transformation produces a "wavy" effect in the metric, even though the underlying geometry remains flat.

Let us consider a completely flat spacetime described in a coordinate system $\overline{x} = (\overline{x}_0, \overline{x}_1, \overline{x}_2, \overline{x}_3)$, where the metric is simply the Minkowski metric $\eta_{\mu\nu}$ (using the East Coast convention).

Now, we define a new coordinate system x^μ that differs only slightly from $\overline{x}^\mu$:

- $x_0 = \overline{x}_0$
- $x_1 = \overline{x}_1$
- $x_2 = \overline{x}_2$
- $x_3 = \overline{x}_3 + \frac{a}{\omega} \cos\left[\omega(x_0 - x_3)\right]$

We now compute the line element in this new coordinate system. Since the transformation is small (of order a), we expand the metric to first order:

$$\begin{aligned} ds^2 = d\overline{s}^2 &= -d\overline{x}_0^2 + d\overline{x}_1^2 + d\overline{x}_2^2 + d\overline{x}_3^2 \\ &= -dx_0^2 + dx_1^2 + dx_2^2 + \left[dx_3 + a\sin\left(\omega(x_0 - x_3)\right)(dx_0 - dx_3)\right]^2 \\ &= -dx_0^2 + dx_1^2 + dx_2^2 + \left[1 - 2a\sin\left(\omega(x_0 - x_3)\right)\right] dx_3^2 + 2a\sin\left(\omega(x_0 - x_3)\right) dx_0 dx_3 + \mathcal{O}(a^2) \end{aligned} \tag{20}$$

This expression shows that the metric in the new coordinates takes the form:

$$g_{\mu\nu} = \eta_{\mu\nu} + a h_{\mu\nu}$$

with the perturbation tensor:

$$h_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & \sin\left[\omega(x_0 - x_3)\right] \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \sin\left[\omega(x_0 - x_3)\right] & 0 & 0 & -2\sin\left[\omega(x_0 - x_3)\right] \end{pmatrix}$$

Note that $h_{\mu\nu}$ is symmetric, as expected.

Despite the appearance of a wave-like perturbation in the metric, this is not a genuine gravitational wave. The spacetime is still flat—no curvature has been introduced. The apparent "wave" is purely a result of the coordinate transformation. This example illustrates how gauge freedom in General Relativity can produce misleading effects: a metric that looks like it contains gravitational waves, even though the geometry is unchanged.

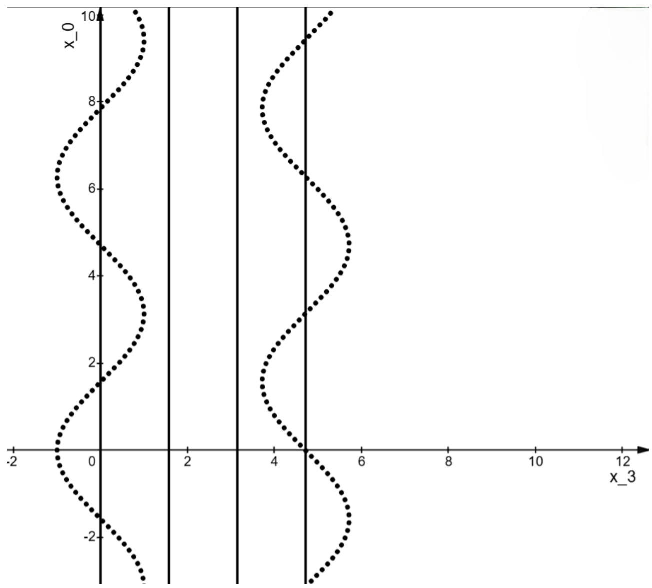


Figure 1: The dashed lines correspond to $x_3 = 0$ (first) and $x_3 = \frac{3\pi}{2}$ (second). The straight vertical lines correspond to points of constant $\overline{x}_3$.

3 Gravitational Wave Detection

The direct detection of gravitational waves represents one of the most formidable experimental challenges in modern physics, requiring measurement precision at the attometer scale. The detection of strain amplitudes on the order of 10^{-21} necessitates instrumentation capable of resolving differential displacements of approximately 10^{-18} m over kilometer-scale baselines—a fractional length change three orders of magnitude smaller than the classical proton diameter. Achieving such sensitivity demands the integration of high-power stabilized laser systems, multi-stage seismic isolation, ultra-high vacuum infrastructure, and advanced quantum-noise-limited interferometry.

This section examines the fundamental operating principles of laser interferometric gravitational-wave observatories, with particular emphasis on the noise limitations inherent in terrestrial detection.

Sensitivity bandwidth fundamentally constrains the astrophysical reach of any gravitational-wave detector. Current-generation terrestrial interferometers, such as Advanced LIGO and Advanced Virgo, operate within the $\sim$10–2000 Hz band, optimized to study the merger phase in compact binary systems involving neutron stars and stellar-mass black holes. Future third-generation observatories like the Einstein Telescope will extend sensitivity down to $\sim$2 Hz, permitting longer-duration observation of inspiral phases and access to higher-mass systems. The space-based Laser Interferometer Space Antenna (LISA) will operate in the 10^{-4}–10^{-1} Hz band, enabling observations of supermassive black hole mergers, extreme mass-ratio inspirals, and galactic binaries that are inaccessible from terrestrial environments.

The following analysis focuses on the operational architecture of ground-based laser interferometers, with specific reference to the Advanced Virgo (AdV) configuration.

3.1 Operating Principles of Ground-Based Laser Interferometers

The foundational architecture of modern gravitational-wave detectors is the dual-recycled Michelson[3] interferometer with Fabry–Pérot arm cavities. A high-power, frequency-stabilized laser source is incident upon a beam splitter, dividing the optical power into two orthogonal arms. Each arm contains a linear Fabry–Pérot resonator formed by an input test mass (IM) and end test mass (EM), enhancing the effective arm length through multipass light storage. The interferometer is maintained at a dark fringe operating point, where constructive interference at the beam splitter directs the carrier field back toward the laser, and the output port is dominated by differential phase signals.

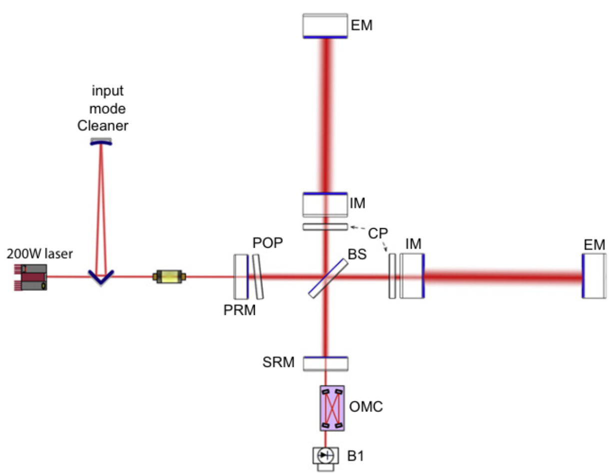


Figure 2: Simplified optical layout of the AdV ITF. Each 3 km long arm cavity isformed by an IM and an EM. The recycling cavities at the center of the interferometer are 12meters long and are formed by the PRM, the SRM and the two IM.

A gravitational wave incident on the detector induces a differential length variation ΔL between the arms. This perturbation phase-modulates the circulating carrier field, generating sidebands that are resonant in the signal recycling

[3]It's a classical Michelson interferometr in which both the laser power and the GW signal are recycled.

cavity and transmitted to the output photodetector. The resulting photocurrent thus encodes the gravitational waveform.

The practical realization of this measurement requires suppression of noise sources several orders of magnitude above the astrophysical signal. The Advanced Virgo design incorporates multiple subsystems to achieve this:

- **Quantum Noise Limitations:** The sensitivity ceiling across much of the detection band is set by the combined effects of shot noise (high frequencies) and radiation pressure noise (low frequencies)[4]. "AdV employs high-power (200W, capable of injecting around 60W after losses) lasers. The power recycling cavity amplifies the circulating power to $\sim$100 kW, reducing shot noise uncertainty. The signal recycling cavity is optically tuned to enhance specific signal sidebands, providing a degree of quantum noise reshaping.

- **Seismic Isolation and Suspension Systems:** The mirrors at the ends of each arm are made from ultra-pure fused silica to minimise internal thermal noise, these test masses are suspended as multi-stage pendulums, providing passive seismic isolation via their transfer function's steep high-frequency roll-off [5]. The AdV Superattenuator system combines pendulums, geometric anti-spring filters, and a final monolithic silica suspension stage to achieve vibration attenuation exceeding 10^{-10} above 10 Hz. Active inertial feedback systems further damp low-frequency resonances.

- **Ultra-High Vacuum System:** The beam tubes and optical chambers are maintained at pressures below 10^{-9} mbar to minimize phase noise from residual gas scattering and refractive index fluctuations.

- **Optical Configuration:** The dual-recycled Fabry–Pérot Michelson topology provides independent control of the circulating power (via the transmissivity of the power recycling mirror) and the signal gain (via the detuning of the signal recycling mirror). This enables optimization of the sensitivity profile for different astrophysical targets[6].

- **Control and Calibration:** The interferometer is maintained at its operating point via a hierarchy of feedback loops employing wavelength-scale precision. Differential arm length control is maintained via radiation pressure actuators and electrostatic drivers.

- **Data Quality and Glitch Mitigation:** Non-Gaussian noise transients (glitches) arising from environmental and instrumental disturbances are

[4] *Shot noise* arises from the discrete nature of photons, producing statistical fluctuations in the detected light intensity and limiting phase measurement precision. *Radiation pressure noise* originates from fluctuations in photon momentum transfer to the mirrors, causing tiny displacements of the test masses. Increasing optical power reduces shot noise but increases radiation pressure noise, leading to a trade-off known as the Standard Quantum Limit.

[5] This feature is explained in grater detail in section 3.2.

[6] By acting on the resonance of the sidebands.

identified and characterized using machine learning algorithms. These methods enable noise regression and data quality vetoes, reducing false alarm rates in astrophysical searches.

The synergistic operation of these subsystems enables strain sensitivities within the most sensitive frequency band (approximately 100–300 Hz) that are sufficient for the routine detection of gravitational waves from coalescing compact binaries.

3.2 Example 2: Seismic Isolation System Simulator

To provide a clearer understanding of one of the noise-reduction techniques employed in gravitational-wave detectors, this section examines the seismic isolation system. Particular attention is given to the pendulum-based suspension design, which is widely used to mitigate ground-induced vibrations. By simulating the behaviour of this system, it is possible to analyse how the pendulum's mechanical properties contribute to the attenuation of seismic noise, particularly within the high-frequency range.

The fundamental principle underlying this approach is the exploitation of the intrinsic mechanical low-pass filter properties of a multi-stage pendulum. Such a configuration induces significant damping of ground vibrations transmitted to the suspended components. This method is especially effective in gravitational-wave observatories, as interferometers are often located in or near populated areas where anthropogenic noise is prevalent, in addition to the ever-present seismic background.

In the Virgo interferometer, the seismic isolation system comprises seven independent pendulum assemblies: one for each end mirror (EM), one for each input mirror (IM), one for the power recycling mirror (PRM), one for the signal recycling mirror (SRM), and one for the beam splitter (BS). Each suspension tower consists of a five-stage pendulum (obviously counting the mirror aswell) with a total length of approximately 15 m.

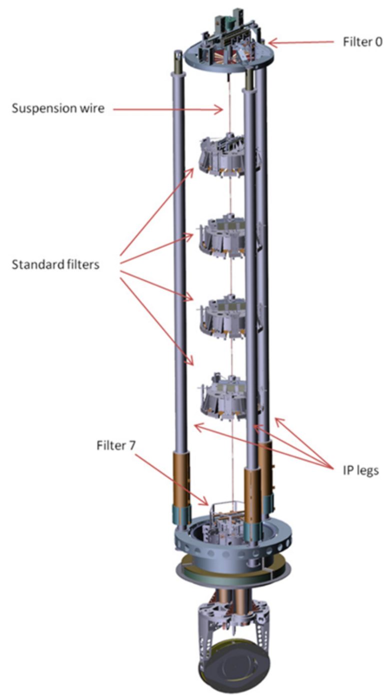


Figure 3: Schematic of the pendulum used in the seismic isolation system.

3.2.1 Derivation of the Model Equations

For the purposes of this simulation, a simplified model is considered, consisting of a two-stage pendulum with identical masses and equal lengths. This reduction in complexity facilitates a more tractable analysis while preserving the essential dynamical characteristics relevant to seismic noise attenuation.

To simulate the noise, ground vibrations parallel to the surface were considered, restricting the analysis to the component aligned with the laser direction. This choice is justified by the fact that the mirrors are sufficiently large and flat to be effectively insensitive to perpendicular displacements. The noise was modeled as a linear combination of a sine wave without phase (corresponding to a pendulum initially at rest) and an exponential damping factor.

The motion of the ceiling was assumed to be absolute and unaffected by the masses of the system. The analysis will firstly be carried out in a reference frame in which the ceiling is at rest. The results will then be transformed back into the laboratory reference frame through a simple Galilean transformation.

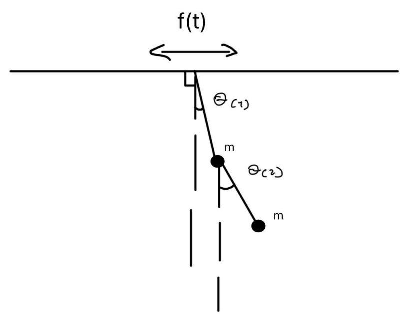


Figure 4: Simulation of the two-stage pendulum response to ground vibrations.

The motion of the ceiling in the laboratory reference frame is described by

$$f(t) = \sum_{i=1}^{n} \left[e^{-\frac{t}{\tau_i}} \sin(\omega_i t) \right] A_i, \tag{21}$$

where each term corresponds to a damped sinusoidal component with amplitude A_i, angular frequency ω_i, and decay constant τ_i.

Its second derivative is

$$f''(t) = \sum_{i=1}^{n} e^{-\frac{t}{\tau_i}} \left[\left(\frac{1}{\tau_i^2} - \omega_i^2 \right) \sin(\omega_i t) - 2\frac{\omega_i}{\tau_i} \cos(\omega_i t) \right] A_i. \tag{22}$$

When moving to the non-inertial frame in which the ceiling is at rest, we must introduce an apparent acceleration along the x-axis:

$$a_p(t) = -f''(t).$$

In this frame, the potential energy of a mass m is

$$U = -m a_p(t)\, x + mgy. \tag{23}$$

—

Kinematics of the two-stage pendulum We now determine the positions and velocities of the two identical masses in order to compute the kinetic energy and to express the potential energy entirely in terms of the angular coordinates θ_1, θ_2 and time t.

First mass. The coordinates of the first mass are

$$\begin{cases} x_1 = \ell \sin\theta_1, \\ y_1 = -\ell\cos\theta_1, \end{cases} \quad \Rightarrow \quad \begin{cases} \dot{x}_1 = \ell\,\dot{\theta}_1\cos\theta_1, \\ \dot{y}_1 = \ell\,\dot{\theta}_1\sin\theta_1. \end{cases}$$

Second mass. The second mass is attached to the first, so its coordinates are

$$\begin{cases} x_2 = x_1 + \ell\sin\theta_2, \\ y_2 = y_1 - \ell\cos\theta_2, \end{cases} \quad \Rightarrow \quad \begin{cases} x_2 = \ell\,[\sin\theta_1 + \sin\theta_2], \\ y_2 = -\ell\,[\cos\theta_1 + \cos\theta_2]. \end{cases}$$

Differentiating with respect to time:

$$\begin{cases} \dot{x}_2 = \ell\,[\dot{\theta}_1\cos\theta_1 + \dot{\theta}_2\cos\theta_2], \\ \dot{y}_2 = \ell\,[\dot{\theta}_1\sin\theta_1 + \dot{\theta}_2\sin\theta_2]. \end{cases}$$

—

Velocities and kinetic energy The squared speeds of the two masses are

$$|\vec{v}_1|^2 = \ell^2\dot{\theta}_1^2,$$

$$|\vec{v}_2|^2 = \ell^2\left[\dot{\theta}_1^2 + \dot{\theta}_2^2 + 2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1-\theta_2)\right].$$

The total kinetic energy of the system is therefore

$$T = \frac{1}{2}m\left(|\vec{v}_1|^2 + |\vec{v}_2|^2\right) = \frac{1}{2}m\ell^2\left[2\dot{\theta}_1^2 + \dot{\theta}_2^2 + 2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1-\theta_2)\right]. \tag{24}$$

—

Potential energy in terms of θ_1, θ_2, and t From the expressions for x_1, y_1, x_2, and y_2, the total potential energy is

$$U(\theta_1,\theta_2,t) = -ma_p(t)\,x_1 + mgy_1 - ma_p(t)\,x_2 + mgy_2. \tag{25}$$

Substituting the coordinates:

$$U(\theta_1,\theta_2,t) = -ma_p(t)\,\ell\,[2\sin\theta_1 + \sin\theta_2] - mg\,\ell\,[2\cos\theta_1 + \cos\theta_2]. \tag{26}$$

This form explicitly shows the dependence of the potential energy on the angular coordinates and on time through $a_p(t)$.

Lagrangian of the system and lagrange equations

$$\mathcal{L}(\theta_1,\theta_2,\dot\theta_1,\dot\theta_2,t) = \frac{1}{2}m\ell^2\left[2\dot\theta_1^2+\dot\theta_2^2+2\dot\theta_1\dot\theta_2\cos(\theta_1-\theta_2)\right] + ma_p(t)\,\ell\,[2\sin\theta_1+\sin\theta_2] + mg\,\ell\,[2\cos\theta_1+\cos\theta_2]. \quad (27)$$

This Lagrangian explicitly depends on time through $a_p(t)$, which is determined by the ceiling motion $f(t)$ (consequentially it is impossible to derive an energy conservation law), and on the generalized coordinates θ_1 and θ_2 along with their time derivatives. At this point we can just use the classical Euler-Lagrange formula for each degree of freedom and optain the following two expressions

$$\theta_2'' = \frac{1}{1-\frac{1}{2}\cos^2(\theta_1-\theta_2)}\Bigg[\theta_2'\theta_1'\sin(\theta_1-\theta_2)+\frac{a_p}{\ell}\cos\theta_2-\frac{g}{\ell}\sin\theta_2-\theta_1'(\theta_1'-\theta_2')\sin(\theta_1-\theta_2) - \frac{\cos(\theta_1-\theta_2)}{2}\left(2\frac{a_p}{\ell}\cos\theta_1-2\frac{g}{\ell}\sin\theta_1-\theta_1'\theta_2'\sin(\theta_1-\theta_2)+\theta_2'(\theta_1'-\theta_2')\sin(\theta_1-\theta_2)\right)\Bigg]$$

$$\theta_1'' = \frac{2\frac{a_p}{\ell}\cos\theta_1-2\frac{g}{\ell}\sin\theta_1-\theta_1'\theta_2'\sin(\theta_1-\theta_2)-\theta_2''\cos(\theta_1-\theta_2)+\theta_2'\,(\theta_1'-\theta_2')\sin(\theta_1-\theta_2)}{2}$$

Due to their high degree of nonlinearity and coupling, these differential equations do not admit a closed-form analytical solution. We therefore proceed by a computational approach in the next section.

3.2.2 Computational solution and results

At this stage, the most suitable approach is to discretize the time interval of interest into a large number of small increments, enabling the system's evolution to be computed step by step. While increasing the number of subdivisions enhances temporal resolution, it also substantially raises the computational cost of the simulation. In particular, higher noise frequencies require a correspondingly greater number of time subdivisions to be accurately resolved. After each interaction, the mirror's position is recalculated in the laboratory reference frame. Beyond these considerations, there is little more to add regarding the simulation methodology; however, the tool developed for this purpose is available upon request.
The following section presents selected results from the simulations conducted.

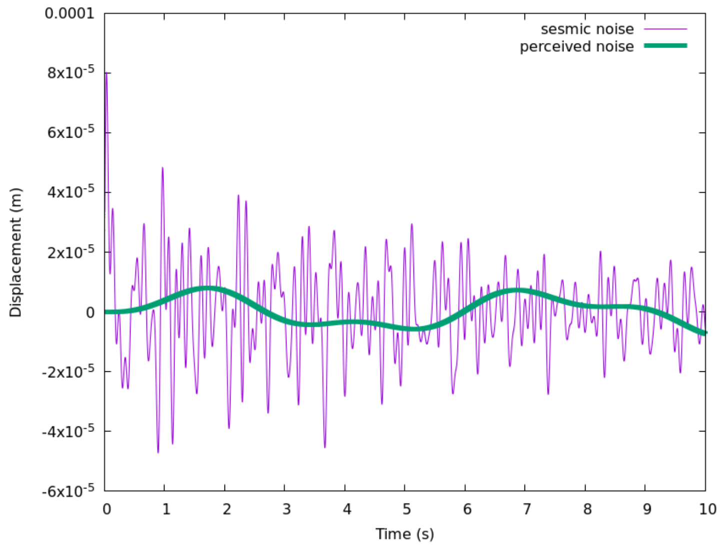


Figure 5: Simulation results for a lower driving frequency, illustrating the system's displacement response over time.

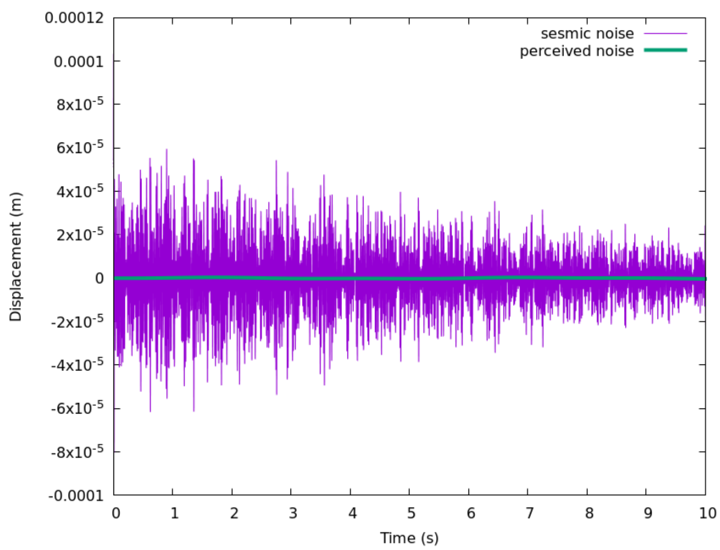


Figure 6: Simulation results for a higher driving frequency, illustrating the system's displacement response over time.

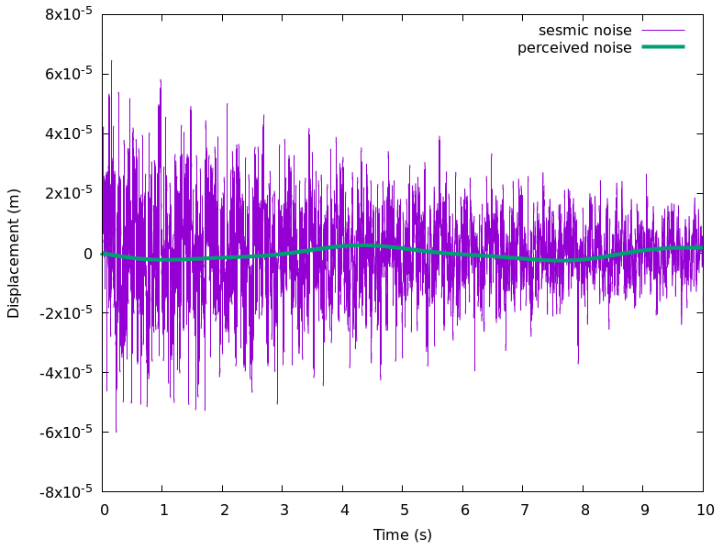


Figure 7: Simulation results for a a mix of lower and higher driving frequency, illustrating the system's displacement response over time.

A Bibliography

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[2] S. Carroll, *Spacetime and Geometry: An Introduction to General Relativity*, Addison-Wesley, 2004.

[3] *Class notes*